

GAZEMM: ROBUST OFF-ANGLE IRIS RECOGNITION VIA GAZE MANIFOLD MATCHING

Yewon Min*

Youngwook Kim*

Jiyoung Park

Moonwook Ryu

Jun Seong Lee

Electronics and Telecommunications Research Institute
Daejeon, Republic of Korea

ABSTRACT

Recent deep learning-based iris recognition methods encode iris images as high-dimensional embeddings and typically rely on point-to-point matching. Under off-angle acquisition, however, gaze-induced appearance changes increase substantial intra-class variation, separating embeddings of the same identity and degrading conventional point-to-point matching. In this paper, we propose GazeMM, an identity-specific gaze manifold-based iris matching framework that explicitly models such gaze-induced variation. Given multi-view enrollment embeddings, GazeMM constructs a low-dimensional affine subspace for each identity and matches a query using its orthogonal residual distance to the subspace, followed by cohort-based local score normalization. Experimental results revealed that GazeMM achieves an average equal error rate of 0.23% under off-angle conditions, while maintaining competitive performance under frontal acquisition. Moreover, it remains effective across different iris feature extractors, demonstrating its generality. Source code for GazeMM is available at <https://github.com/kimyoungwook7/GAZEMM>.

Index Terms— Iris recognition, Off-angle iris, Gaze manifold, Subspace matching, Distance normalization

1. INTRODUCTION

The iris is a highly distinctive and temporally stable biometric trait, making iris recognition one of the most reliable biometric identification modalities [1]. Classical *iris recognition* systems typically rely on iris localization, polar normalization, Gabor-based encoding, and Hamming-distance matching [1, 2, 3, 4], whereas recent deep learning-based approaches learn high-dimensional embeddings directly from input images and compare them using cosine similarity or Euclidean distance [5, 6]. Although both paradigms perform well under controlled acquisition, their performance degrades substantially under off-angle conditions [6, 7, 8, 9]. As the deviation between the camera optical axis and gaze direction increases, geometric distortion and partial occlusion disrupt the correspondence of iris textures [10, 11]. In learned embedding spaces, these effects increase gaze-dependent intra-class variation, thereby making reliable matching more difficult.

To address this challenge, existing approaches to off-angle iris recognition mainly *employ* geometric rectification or robust feature learning. Geometric approaches compensate for off-angle deformation by warping the iris toward a frontal view based on gaze or imaging geometry [7, 8]. Learning-based approaches improve robustness through architectural design or metric learning in the embedding space [5, 6, 12, 13]. However, both strategies primarily aim to suppress gaze-induced variation before matching, leaving structural relationships among multiple same-identity embeddings largely unexploited at the matching stage.

Meanwhile, in related domains such as face and object recognition, subspace-based approaches *have been used to jointly* model multiple observations of an identity as a low-dimensional subspace and perform matching based on their geometric relationship to that subspace [14, 15, 16]. This perspective is grounded in the observation that variations induced by factors such as pose or illumination tend to concentrate along a small number of directions in the feature space, rather than varying independently.

Motivated by this observation, we *posit* that gaze-induced intra-class variation in iris embeddings *can likewise be captured by* a low-dimensional structure. Building on this *premise*, we propose a gaze manifold-based iris matching framework that represents multi-view embeddings of each identity as an identity-specific affine subspace and matches a query using its orthogonal residual distance to that subspace. By reducing the influence of feature variations aligned with the learned intra-class directions, the proposed formulation accounts for gaze-induced embedding changes without requiring explicit gaze estimation or geometric rectification.

The main contributions are threefold:

- We investigate gaze-induced intra-class variations in off-angle iris embeddings and show that it exhibits low-dimensional structure at the matching stage.
- We propose GazeMM, which models multi-view enrollment embeddings as an identity-specific affine manifold and matches queries using cohort-normalized residual distance.
- We validate GazeMM across twelve datasets and multiple feature extractors, showing strong off-angle performance and consistent generalization.

*These authors contributed equally to this work.

2. APPROACH

2.1. Overall Framework

Given a near-infrared (NIR) eye image, quality filtering is first performed, and an embedding is extracted without explicit iris segmentation or polar normalization. In the enrollment stage, multi-gaze embeddings of each user are used to construct an identity-specific affine subspace. During authentication, residual distances between a query and the enrolled subspaces are computed and normalized using cohort-based distance normalization, and the resulting normalized distances are used for final matching.

2.2. Quality Filtering and Embedding Extraction

2.2.1. Quality Filtering

To reject severely blurred or low-information frames, we use the Laplacian variance f as a lightweight quality measure. Considering its right-skewed distribution, the threshold is defined in the log domain using the median and median absolute deviation (MAD):

$$\text{keep} \iff \log f \geq \text{median}(\log f) - c \cdot 1.4826 \cdot \text{MAD}(\log f), \quad (1)$$

where c controls the filtering strictness. The statistics are estimated from the training data and fixed during evaluation.

2.2.2. Input Representation and Embedding Extraction

Given an NIR periocular image, it is padded to a square while preserving its aspect ratio, resized to 224×224 , and replicated across three channels. No explicit iris segmentation or polar normalization is applied. The preprocessed image is then fed into a ResNet-34 [17], to obtain a 512-dimensional embedding, which is L_2 -normalized as

$$x = \frac{f(I)}{\|f(I)\|_2}, \quad (2)$$

where $f(I)$ denotes the network output for input image I , and x is the normalized embedding used for subsequent subspace modeling and matching.

The embedding network is trained using batch-hard triplet learning [18]. For each anchor, the farthest positive (from the same identity) and nearest negative samples (from a different identity) are selected, and the margin-free softplus loss is applied as

$$\ell = \text{softplus}(d(a, p) - d(a, n)) = \log(1 + e^{d(a, p) - d(a, n)}), \quad (3)$$

where a , p , and n denote the anchor, positive, and negative samples, respectively, and $d(\cdot, \cdot)$ is the Euclidean distance between L_2 -normalized embeddings.

2.3. Identity-Specific Affine Subspace Modeling

2.3.1. Multi-View Enrollment Representation

To model gaze-dependent intra-class variation, we represent each identity's multiple enrollment embeddings as a low-dimensional affine subspace. For identity i with N enrollment embeddings $\{x_n\}_{n=1}^N$, the mean embedding and centered feature matrix are

$$\mu_i = \frac{1}{N} \sum_{n=1}^N x_n, A_i = [x_1 - \mu_i, x_2 - \mu_i, \dots, x_N - \mu_i]. \quad (4)$$

The top k eigenvectors of $A_i A_i^\top$ form an orthonormal basis U_i that captures the dominant intra-class variation. The effective subspace dimension is limited to $\min(k, N - 1, D)$, where k is a hyperparameter. The resulting identity-specific affine subspace is represented as $M_i = (\mu_i, U_i)$.

2.3.2. Point-to-Subspace Residual Distance

At the authentication stage, given a query embedding q , it is projected onto the identity-specific affine subspace M_i :

$$p_i = \mu_i + U_i U_i^\top (q - \mu_i), \quad (5)$$

where p_i denotes the orthogonal projection of q onto the subspace. The point-to-subspace residual distance is then defined as

$$d_i = \|q - p_i\|_2 = \|(\mathbf{I}_D - U_i U_i^\top)(q - \mu_i)\|_2, \quad (6)$$

where $\mathbf{I}_D$ denotes the $D \times D$ identity matrix. A smaller d_i indicates greater consistency with the intra-class variation represented by identity i .

2.4. Cohort-Normalized Matching

Residual-distance scale can vary across queries and enrolled subspaces, making a single global threshold suboptimal for authentication. To reduce this variation, we adapt local scaling [19] to cohort-based score normalization. Let d_{ij} denote the residual distance between query j and enrolled subspace M_i . The cohort mean distances are computed as

$$\bar{d}_i = \frac{1}{|C_q|} \sum_{j' \in C_q} d_{ij'}, \quad \bar{d}_j = \frac{1}{|C_M|} \sum_{i' \in C_M} d_{i'j},$$

where C_q and C_M denote the cohort sets of queries and enrolled subspaces, respectively. The normalized score is defined as

$$s_{ij} = \frac{d_{ij}}{\sqrt{\bar{d}_i \bar{d}_j}}.$$

A smaller s_{ij} indicates a stronger match, and authentication is performed by applying a single global threshold.

3. EXPERIMENTAL RESULTS

3.1. Experimental Setup

Dataset: We evaluate on twelve iris datasets: eight frontal datasets (CASIA v1.0, CASIA-Iris-Interval, CASIA-Iris-Thousand, CASIA-Iris-Lamp [20], PolyU Iris DB [21], MMU-v1 [22], and the Cogent and Vista subsets of IIITD Contact Lens Iris DB [23], and four off-angle datasets (Eye-Movement [24], OpenEDS [25], CASIA-Iris-Degradation [20], and CASIA-Iris-Africa [26]). All datasets were obtained through official public or authorized-access channels [under the providers’ license agreements](#). Identities are defined at the eye level, treating the left and right eyes as different classes. Dataset statistics and subset details are provided in the [supplementary material](#).

Comparison methods: We compare against classical methods (WAHET [2], [USITv3](#) [3], and OSIRIS [4]), hybrid methods (HDBIF [27] and WCI [28]), and deep-learning methods (UE-UGCL [12], Maxout-BA [12], ArcIris [13], TripleIris [13], and DGR [29]). All methods are evaluated using the authors’ released implementations and recommended or default configurations. Publicly released pretrained models are used without retraining. The exceptions are UE-UGCL and Maxout-BA, for which trained weights are unavailable; these methods are trained on the same training folds as our method, without the external-data pretraining used in their original settings. Detailed configurations are provided in the [supplementary material](#).

Evaluation metrics: We report equal error rate (EER), false rejection rate (FRR) at a false acceptance rate of 10^{-3} , Rank-1 accuracy, and failure-to-enroll rate (FTE), all in %. Samples that fail to enroll are excluded from the matching pool; thus, the reported performance of methods with high FTE may be optimistic.

Evaluation protocol: We use subject-disjoint 2-fold cross-validation with three fixed random seeds. The proposed method, UE-UGCL, and Maxout-BA are trained on each training fold, with 15% of the training identities used for validation and the best checkpoint selected by lowest validation EER for testing. The remaining methods are evaluated on the same test folds. The proposed method uses leave-one-out identity-specific affine subspaces for matching, while comparison methods follow their original matching protocols.

Implementation details: We use an ImageNet-pretrained ResNet34 with a 512-dimensional embedding and train it using the soft-margin batch-hard triplet loss with Adagrad. We set the quality-filtering parameter to $c=2.0$ and the subspace dimension to $k=3$. Further training details and hyperparameter analyses are provided in the [supplementary material](#).

Table 1: Performance comparison averaged over four off-angle and eight frontal datasets (%). Baselines are grouped as classical, hybrid, and deep-learning-based methods. Unless otherwise noted, **bold** and underlined values denote the best and second-best results, respectively.

Method	Off-angle (avg.)				Frontal (avg.)			
	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE
WAHET [2]	35.83	95.28	10.60	0.0	9.28	33.02	90.17	0.0
USITv3 [3]	33.23	81.71	50.11	0.0	11.12	20.22	87.34	0.0
OSIRIS [4]	24.18	82.92	62.06	9.7	4.62	11.67	97.45	1.4
HDBIF [27]	26.25	68.27	77.05	0.0	1.41	2.88	99.78	0.0
WCI [28]	<u>7.10</u>	<u>45.70</u>	<u>87.07</u>	39.6	0.17	0.22	99.94	16.6
DGR [29]	17.63	72.59	83.96	20.4	3.18	12.26	99.23	1.4
UE-UGCL [12]	11.98	66.29	84.10	20.4	10.48	50.99	89.95	1.4
Maxout-BA [12]	22.13	81.14	79.29	20.4	15.90	66.05	82.69	1.4
ArcIris [13]	16.51	59.42	83.97	1.2	0.64	<u>1.16</u>	99.84	0.0
TripleIris [13]	15.62	82.13	81.96	0.0	0.83	3.14	99.79	0.0
GazeMM (Ours)	0.23	8.18	94.12	2.8	<u>0.34</u>	1.87	<u>99.87</u>	2.6

Table 2: EER and FTE on four off-angle datasets (%): Eye-Movement (EM), OpenEDS (OED), CASIA-Iris-Degradation (Deg), and CASIA-Iris-Africa (Afr).

Method	EM		OED		Deg		Afr	
	EER	FTE	EER	FTE	EER	FTE	EER	FTE
WAHET [2]	35.98	0.0	38.32	0.0	29.23	0.0	39.80	0.0
USITv3 [3]	33.14	0.0	37.59	0.0	25.56	0.0	36.66	0.0
OSIRIS [4]	22.94	32.8	23.27	1.8	27.30	1.3	23.21	3.0
HDBIF [27]	27.35	0.0	28.49	0.0	26.61	0.0	22.57	0.0
WCI [28]	12.82	39.3	<u>6.27</u>	26.3	<u>5.06</u>	25.1	<u>4.25</u>	67.7
DGR [29]	14.10	39.1	15.40	25.4	17.26	13.8	23.79	3.3
UE-UGCL [12]	<u>10.70</u>	39.1	11.61	25.4	13.16	13.8	12.45	3.3
Maxout-BA [12]	15.10	39.1	27.62	25.4	18.64	13.8	27.16	3.3
ArcIris [13]	15.64	2.8	16.20	0.1	16.54	1.6	17.69	0.1
TripleIris [13]	14.94	0.0	11.98	0.0	17.98	0.0	17.60	0.0
GazeMM (Ours)	0.00	1.5	0.57	0.3	0.25	3.0	0.10	6.3

3.2. Quantitative Results

Table 1 compares the average performance over four off-angle and eight frontal datasets. On the off-angle datasets, the proposed method achieves the best overall performance, substantially outperforming the second-best WCI while maintaining a low FTE. On the frontal datasets, although ranking second overall, it remains competitive with the best WCI while achieving a substantially lower FTE. Overall, the proposed method provides substantial gains under off-angle without compromising performance under frontal acquisition.

Breaking down the average off-angle performance by dataset, Table 2 shows that the proposed method achieves the best EER on all four datasets. In addition, unlike several comparison methods that exhibit high FTE, the proposed method maintains consistently low FTE across all datasets.

3.3. Generalization Across Feature Extractors

To evaluate the generality of GazeMM, we apply it to several existing iris embeddings, averaging over four off-angle

Table 3: Comparison of matching schemes per feature extractor averaged over four off-angle datasets (%).

Feature extractor	Point-to-point			Proposed matching (Ours)	
	EER	FRR		EER	FRR
UE-UGCL [12]	<u>11.98</u>	66.29	→	0.67	14.16
Maxout-BA [12]	22.13	81.14	→	1.50	21.39
ArcIris [13]	16.51	<u>59.42</u>	→	<u>0.49</u>	<u>13.35</u>
TripletIris [13]	15.62	82.13	→	0.78	15.09
ResNet34 (Ours)	2.50	32.71	→	0.23	8.18

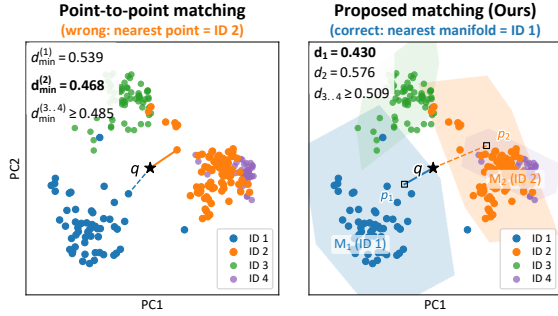


Fig. 1: Point-to-point (left) vs. proposed (right) matching for a query q (star) on CASIA-Iris-Degradation, shown in a PCA projection. Colors denote identities, shaded regions the manifolds M_i , squares the projections p_i . The proposed manifold matching method correctly matches the genuine identity (ID 1), unlike point-to-point matching.

datasets. As shown in Table 3, the proposed matching consistently reduces both EER and FRR over conventional point-to-point method across all feature extractors. Because GazeMM focuses on matching strategy rather than feature extraction, it adopts a simple ResNet34 encoder while still achieving low error rates. These results demonstrate that GazeMM generalizes across diverse iris embeddings without architectural modification. Per-dataset results are provided in the [supplementary material](#).

3.4. Qualitative Results

Fig. 1 qualitatively illustrates how the proposed manifold matching handles gaze-induced intra-class variation. For the same query q from ID 1, point-to-point matching is misled by a nearby embedding from ID 2, whereas the proposed method correctly matches q to M_1 using residual distance. This highlights the advantage of modeling identity-specific variation rather than relying on individual enrollment points.

Fig. 2 further illustrates the effect of the proposed manifold matching by comparing the genuine and impostor distance distributions of point-to-point and point-to-manifold matching. Under off-angle settings, same-identity queries can remain distant from individual enrollment points, increasing the genuine distance. By modeling gaze-induced variation as

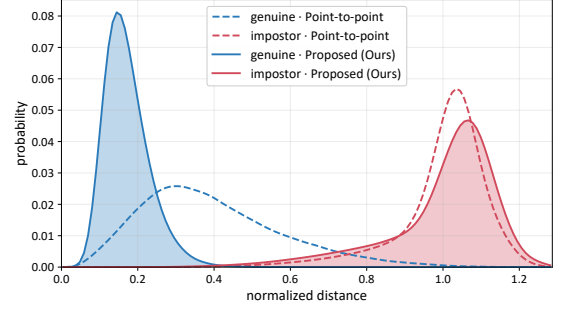


Fig. 2: Genuine and impostor distance distributions of point-to-point vs. proposed manifold matching with the same ResNet34-based embedding on CASIA-Iris-Degradation.

Table 4: Component-wise ablation study on CASIA-Iris-Degradation (%).

Variant	Manifold matching	Score norm.	Quality filtering	EER	FRR
Point-to-point matching		✓	✓	4.98	45.41
w/o Score normalization	✓		✓	0.60	4.49
w/o Quality filtering	✓	✓		0.29	2.00
GazeMM (Ours)	✓	✓	✓	0.25	1.23

a subspace, manifold matching reduces this distance. As a result, the genuine distribution shifts toward smaller distances and becomes more compact, while the impostor distribution remains relatively similar, leading to better separation between the two. This suggests that the proposed matching primarily alleviates intra-class distance variation rather than uniformly shrinking all distances.

3.5. Ablation Study

Table 4 compares the effects of the major components of GazeMM on CASIA-Iris-Degradation. Replacing manifold matching with point-to-point matching causes the largest performance degradation, confirming that manifold matching is the dominant component of GazeMM. Score normalization provides further gains, while the quality filtering also improves performance by filtering low-quality inputs.

4. CONCLUSION

This paper proposed GazeMM for robust off-angle iris recognition. Under off-angle acquisition, embeddings of the same identity can vary substantially with gaze direction, limiting conventional point-to-point matching. To address this limitation, GazeMM models multi-view embeddings of the same identity as an identity-specific affine subspace and performs matching based on the normalized residual distance between a query and the subspace. Experiments across twelve datasets show that GazeMM maintains competitive performance under

frontal conditions while achieving an average EER of 0.23% under off-angle conditions, substantially outperforming existing methods. In addition, experiments with existing iris embeddings demonstrate that the proposed matching remains effective when paired with different feature extractors, allowing it to be readily incorporated into existing deep iris recognition models without retraining or architectural changes.

5. ACKNOWLEDGMENTS

This work was supported by an XXXX grant funded by XXX XXXXXXX XXXXXXXXXX (No. XXXXXXXXXX). The authors used Claude (Anthropic) and ChatGPT (OpenAI) to assist with English translation and editing of the manuscript text. All content was reviewed, verified, and revised by the authors.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data from publicly released iris datasets [20, 21, 22, 23, 24, 25, 26], obtained under their respective license agreements. Ethical approval was not required as confirmed by the licenses attached to the datasets.

7. REFERENCES

- [1] J. Daugman, "How iris recognition works," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 14, no. 1, pp. 21–30, Jan. 2004.
- [2] A. Uhl and P. Wild, "Weighted adaptive Hough and ellipsoidal transforms for real-time iris segmentation," in *Proc. IAPR Int. Conf. Biometrics (ICB)*, 2012, pp. 283–290.
- [3] C. Rathgeb, A. Uhl, P. Wild, and H. Hofbauer, "Design decisions for an iris recognition SDK," in *Handbook of Iris Recognition*, pp. 359–396. Springer, 2nd edition, 2016.
- [4] N. Othman, B. Dorizzi, and S. Garcia-Salicetti, "OSIRIS: An open source iris recognition software," *Pattern Recognit. Lett.*, vol. 82, pp. 124–131, 2016.
- [5] Z. Zhao and A. Kumar, "Towards more accurate iris recognition using deeply learned spatially corresponding features," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 3829–3838.
- [6] E. Jalilian, G. Wimmer, A. Uhl, and M. Karakaya, "Deep learning based off-angle iris recognition," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2022, pp. 4048–4052.
- [7] S. A. C. Schuckers, N. A. Schmid, A. Abhyankar, V. Dorairaj, C. K. Boyce, and L. A. Hornak, "On techniques for angle compensation in nonideal iris recognition," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 37, no. 5, pp. 1176–1190, Oct. 2007.
- [8] E. Jalilian, M. Karakaya, and A. Uhl, "CNN-based off-angle iris segmentation and recognition," *IET Biometrics*, vol. 10, no. 5, pp. 518–535, 2021.
- [9] Y. Mi et al., "ImmerIris: A large-scale dataset and benchmark for off-axis and unconstrained iris recognition in immersive applications," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026.
- [10] J. Thompson, P. Flynn, C. Boehnen, and H. Santos-Villalobos, "Assessing the impact of corneal refraction and iris tissue non-planarity on iris recognition," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 8, pp. 2102–2112, Aug. 2019.
- [11] M. Karakaya, D. Barstow, H. Santos-Villalobos, and J. Thompson, "Limbus impact on off-angle iris degradation," in *Proc. Int. Conf. Biometrics (ICB)*, 2013, pp. 1–6.
- [12] J. Wei, H. Huang, Y. Wang, R. He, and Z. Sun, "Towards more discriminative and robust iris recognition by learning uncertain factors," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 865–879, 2022.
- [13] S. K. Khan, P. J. Flynn, and A. Czajka, "Lowering the barrier to IREX participation: Open-source algorithms, toolkit, and benchmarking for iris recognition," arXiv:2605.20735, 2026.
- [14] O. Yamaguchi, K. Fukui, and K. Maeda, "Face recognition using temporal image sequence," in *Proc. IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, 1998, pp. 318–323.
- [15] H. Cevikalp and B. Triggs, "Face recognition based on image sets," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2010, pp. 2567–2573.
- [16] L. S. Souza, N. Sogi, B. B. Gatto, T. Kobayashi, and K. Fukui, "Grassmannian learning mutual subspace method for image set recognition," *Neurocomputing*, vol. 517, pp. 20–33, 2023.
- [17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [18] A. Hermans, L. Beyer, and B. Leibe, "In defense of the triplet loss for person re-identification," arXiv:1703.07737, 2017.
- [19] L. Zelnik-Manor and P. Perona, "Self-tuning spectral clustering," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2004, pp. 1601–1608.
- [20] CASIA, "CASIA iris image databases (V1.0, V4, and Degradation V1.0)," [Online]. Available: <http://biometrics.idealtest.org>.
- [21] A. Kumar and A. Passi, "Comparison and combination of iris matchers for reliable personal authentication," *Pattern Recognit.*, vol. 43, no. 3, pp. 1016–1026, 2010.
- [22] Multimedia University, "MMU iris image database," 2004. [Online]. Available: <http://pesona.mmu.edu.my>.
- [23] N. Kohli, D. Yadav, M. Vatsa, and R. Singh, "Revisiting iris recognition with color cosmetic contact lenses," in *Proc. Int. Conf. Biometrics (ICB)*, 2013, pp. 1–7.
- [24] National Information Society Agency (NIA), "Eye movement video data (AI-Hub)," Dataset no. 548, 2022. [Online]. Available: <https://aihub.or.kr>.
- [25] S. J. Garbin, Y. Shen, I. Schuetz, R. Cavin, G. Hughes, and S. S. Talathi, "OpenEDS: Open eye dataset," arXiv:1905.03702, 2019.
- [26] J. Muhammad, Y. Wang, J. Hu, K. Zhang, and Z. Sun, "CASIA-Iris-Africa: A large-scale African iris image database," *Mach. Intell. Res.*, vol. 21, no. 2, pp. 383–399, 2024.

- [27] A. Czajka, D. Moreira, K. Bowyer, and P. Flynn, “Domain-specific human-inspired binarized statistical image features for iris recognition,” in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2019, pp. 959–967.
- [28] Worldcoin, “Open IRIS: Iris recognition inference system,” 2023. [Online]. Available: <https://github.com/worldcoin/open-iris>.
- [29] M. Ren, Y. Wang, Z. Sun, and T. Tan, “Dynamic graph representation for occlusion handling in biometrics,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2020, pp. 11940–11947.